Comprehensive Research Report: Precipitation Hardening Datasets and Mechanical Property Mappings in the Al-Mg-Si (6xxx) Ternary System

Executive Summary

The optimization of Aluminium-Magnesium-Silicon (Al-Mg-Si) alloys, commercially designated as the 6xxx series, constitutes a central pillar of modern materials engineering, particularly within the automotive, aerospace, and construction sectors. These alloys are prized for their exceptional strength-to-weight ratios, corrosion resistance, and formability. However, their mechanical performance—specifically Yield Strength (YS), Ultimate Tensile Strength (UTS), Shear Strength, and Hardness—is not intrinsic to their composition alone but is strictly governed by the kinetics of precipitation hardening. The transformation of a supersaturated solid solution (SSSS) into a complex hierarchy of metastable nanoscale precipitates (clusters, GP zones, β'', β', Q') defines the material's final properties.

This report provides an exhaustive analysis of the available data ecosystems that map these compositional and thermal variables to mechanical outcomes. Moving beyond traditional empirical handbooks, this research identifies and deconstructs high-dimensional datasets suitable for Integrated Computational Materials Engineering (ICME) and Machine Learning (ML) applications. Key resources analyzed include the "1154 Alloys" dataset for unsupervised learning, granular kinetic datasets for AA6014 and AA6056 detailing time-temperature-property (TTP) trajectories, and ab initio computational datasets derived from Density Functional Theory (DFT).

The analysis reveals that while static property data (e.g., T6 peak strength) is abundant, the critical value lies in *kinetic* datasets that capture the path-dependence of strengthening. Specifically, the data highlights the profound influence of "thermal history"—such as natural aging (NA) intervals and pre-aging (PA) treatments—on the final bake-hardening response. Furthermore, this report addresses the scarcity of direct Shear Yield Strength data, proposing robust correlations with Shear Modulus (G) derived from first-principles calculations. By synthesizing experimental data with theoretical metallurgy, this document establishes a unified framework for utilizing existing datasets to predict and optimize the mechanical response of the Al-Mg-Si ternary system.

---

1. Introduction: The Paradigm Shift in Light Alloy Development

1.1 The Industrial Imperative of Al-Mg-Si Alloys

The global drive toward lightweighting, mandated by fuel efficiency standards and carbon emission targets, has placed the Al-Mg-Si (6xxx series) alloy family at the forefront of structural material design. Unlike the non-heat-treatable 5xxx (Al-Mg) series or the high-strength but heavy 7xxx (Al-Zn-Mg) series, 6xxx alloys offer a unique compromise: they are soft and formable in the solution-treated state (T4), allowing for complex stamping or extrusion, yet can be strengthened significantly through a subsequent artificial aging heat treatment (T6).

This "age hardenability" is the functional property that data scientists and metallurgists seek to model. It allows an automotive body panel to be stamped easily and then harden during the paint-bake cycle (typically 180°C for 20-30 minutes). Consequently, the datasets of highest value are not merely static lists of properties but dynamic maps of how strength evolves as a function of time (t) and temperature (T).

1.2 From Trial-and-Error to Data-Driven Metallurgy

Historically, alloy development followed a "hill-climbing" approach: metallurgists would incrementally vary the Magnesium (Mg) or Silicon (Si) content, conduct tensile tests, and plot the results. This generated vast amounts of isolated analog data, much of which remains locked in paper archives. The modern era of Materials Informatics demands digital, structured datasets that can train Machine Learning (ML) algorithms.

The user's query targets this precise intersection: the need for structured data that links **Composition** (Al, Mg, Si, Cu, etc.) + **Process** (Aging Temp, Time) $\rightarrow$ **Properties** (YS, UTS, Hardness). This report curates such datasets from the open literature, analyzing their structure, fidelity, and metallurgical implications. It bridges the gap between raw CSV files found in repositories like Kaggle and the deep physical metallurgy described in peer-reviewed literature.

1.3 Scope and Structure of the Report

This document is structured to serve as a comprehensive reference for professional peers.

- **Section 2** establishes the physical metallurgy framework, defining the phases and mechanisms that the data represents.
- **Section 3** provides a detailed review of the "Big Data" sources, specifically the 1154-alloy dataset.
- **Section 4** and **Section 5** conduct deep-dive analyses into specific kinetic datasets for

AA6056 and AA6014, reconstructing the data tables and interpreting the trends.
- **Section 6** addresses the computational aspect, reviewing DFT and ML-generated data.
- **Section 7** synthesizes the findings, addressing specific mechanical properties like Shear Yield Strength and detailing the limitations of current data availability.

---

2. The Al-Mg-Si Ternary System: Metallurgical Basis for Data Interpretation

To correctly interpret or utilize a dataset concerning Al-Mg-Si alloys, one must possess a nuanced understanding of the underlying physical phenomena. A data point representing "Hardness at 180°C" is actually a proxy measurement for the number density and aspect ratio of nanometer-scale precipitates.

2.1 The Precipitation Sequence

The mechanical data found in the identified datasets is the macroscopic result of a specific decomposition sequence of the Supersaturated Solid Solution (SSSS). Upon quenching from the solution heat treatment temperature (typically 530°C–570°C), the alloy is in a high-energy unstable state. The relaxation of this state follows a generally accepted pathway [1]:

$$SSSS \rightarrow \text{ Atomic Clusters (Mg/Si)} \rightarrow \text{ GP Zones} \rightarrow \beta'' \rightarrow \beta', U1, U2,$$

2.1.1 Atomic Clusters and the "Negative Effect"

Data regarding "Natural Aging" (T4) refers to the formation of solute clusters and Guinier-Preston (GP) zones at room temperature.

- **Data Implication:** These clusters contribute to yield strength (T4 strength is higher than as-quenched strength) but can be detrimental to T6 properties.
- **Mechanism:** Clusters formed during long room-temperature storage are often stable. When the alloy is subsequently heated to artificial aging temperatures (e.g., 170°C), these clusters do not act as nuclei for the strengthening phase (β'') but instead dissolve or coarsen, depleting the matrix of solute. This results in a "dip" in the aging curve or a lower peak hardness, a phenomenon clearly visible in the kinetic datasets discussed in Section 5.

2.1.2 The β'' Phase (Peak Aging)

The β'' ($\mathrm{Mg_5Si_6}$) phase is the primary hardening agent. It forms fine, coherent needles along the $\langle 100 \rangle_{\mathrm{Al}}$ directions.

- **Data Implication:** The "Peak Aged" (T6) data points in any dataset correspond to the maximum volume fraction of these fine needles. The yield strength here is governed by the Orowan looping mechanism or precipitate shearing, depending on the needle radius.

2.1.3 The Over-Aged Phases (β', Q')

As aging continues or temperature increases (e.g., >200°C), β'' transforms into rod-like β' or lath-like Q' (in Cu-containing alloys).

- **Data Implication:** This microstructural coarsening manifests in the data as a gradual decline in Hardness and YS, usually accompanied by an increase in ductility (Elongation).

2.2 Compositional Variables and the "Alloy Genome"

The "composition" columns in datasets are not merely labels; they define the thermodynamic potential of the system.

2.2.1 The Mg/Si Ratio

The equilibrium phase $\mathrm{Mg_2Si}$ has a mass ratio of Mg:Si $\approx$ 1.73.

- **Balanced Alloys:** Alloys near this ratio maximize the volume fraction of $\mathrm{Mg_2Si}$ for a given total solute content.
- **Excess Silicon:** Most commercial high-strength alloys (like AA6082, AA6014) contain "excess Si" (Si > Mg/1.73).
  - **Effect:** Excess Si promotes the nucleation of β'', refining the precipitate size and increasing strength. Datasets often show a positive correlation between Si content and Peak Hardness.[4]
  - **Mechanism:** Si clusters act as heterogeneous nucleation sites. However, excess Si can also precipitate as elemental Silicon in grain boundaries, which may reduce toughness—a trade-off visible in "Elongation" columns of datasets.

2.2.2 The Role of Copper (Cu)

While the user queried the ternary Al-Mg-Si system, many relevant datasets include Quaternary Al-Mg-Si-Cu alloys (e.g., AA6056, AA6013, AA6061).

- **Data Impact:** Cu completely alters the precipitation sequence, introducing the Q phase family ($\mathrm{AlCuMgSi}$).
- **Observation:** In datasets, Cu-containing alloys exhibit a "broader" aging peak

(resistance to over-aging) and higher absolute strength values compared to Cu-free ternary alloys.[5]

3. High-Dimensional Datasets: The "1154 Alloys" Compendium

In the realm of unsupervised machine learning and broad alloy classification, one dataset stands out for its scale and comprehensiveness. This resource is essential for any researcher attempting to map the global landscape of aluminium alloys before diving into specific kinetic studies.

3.1 Dataset Overview and Access

The dataset is associated with the study *"Unsupervised machine learning discovers classes in aluminium alloys"* by Bhat, Barnard, and Birbilis.[7]

- **Repository:** Mendeley Data
- **DOI:** 10.17632/tvtg7gs59p.1
- **Format:** Structured tabular data (CSV/Excel).
- **Scale:** 1,154 unique alloy instances.

3.2 Feature Engineering: Columns and Structure

This dataset is particularly valuable because it does not rely solely on compositional percentages. It integrates "Engineering Descriptors" (processing history) which are critical for Al-Mg-Si alloys.

3.2.1 Compositional Features (Chemical Descriptors)

The dataset includes specific concentrations (in wt.%) for a wide array of elements, defining the vector space of the alloy's chemistry:

- **Major Solutes:** Al, Mg, Si, Cu, Zn.
- **Dispersoid Formers:** Mn, Cr, Zr (critical for grain structure control).
- **Impurities:** Fe (detrimental to ductility).
- **Trace Elements:** Ti, B, Sc, Li, Ag.

This granularity allows for the analysis of subtle effects, such as the influence of Ag (Silver) on stimulating precipitation in high-performance 6xxx alloys.[9]

3.2.2 Processing Features (Engineering Descriptors)

Recognizing that composition alone does not define an alloy's state, the dataset encodes the **Processing Type**.[8] This is a categorical variable mapped to standard temper designations:

- **SHT:** Solution Heat Treated.

- **T1/T4:** Naturally Aged (mapping to cluster formation).
- **T5:** Artificially Aged from hot working temperature (common for extrusions like AA6063).
- **T6:** Solutionized + Artificially Peak Aged (the standard reference for max strength).
- **T7:** Over-aged (stabilized).
- **H-Series:** Strain hardened (cold work).

Insight for the User: This column is vital. A model trained without differentiating between "T4" and "T6" would fail to predict properties accurately, as the same composition can exhibit yield strengths differing by 100% depending on this processing flag.

3.3 Target Variables (Mechanical Properties)

The dataset includes the three primary mechanical metrics requested by the user:

1. **Tensile Strength (UTS):** Range 44–820 MPa.
2. **Yield Strength (YS):** Range 151–790 MPa.
3. **Elongation (%):** Range 0.5–50%.

While **Shear Strength** is not an explicit column, the correlation between Tensile Yield Strength (σ_y) and Shear Yield Strength (τ_y) is robust for isotropic polycrystals (typically $\tau_y \approx 0.577\sigma_y$ via von Mises criterion).

3.4 Insights from Unsupervised Analysis

The authors utilized Iterative Label Spreading (ILS) to identify 8 distinct classes of alloys.[8]

- **Class 3:** Identified as "Cold-worked and Artificially Aged" alloys.
- **Class 6:** "Naturally Aged" alloys.
- **Relevance:** This clustering confirms that the dataset contains sufficient signal to distinguish between the different metallurgical states of Al-Mg-Si alloys purely based on data patterns. It validates the dataset's quality for training predictive regression models for YS and Hardness.

---

4. Kinetic Dataset Case Study: AA6056 (Al-Mg-Si-Cu)

While the "1154 Alloys" dataset provides breadth, it lacks the temporal resolution to describe *aging curves*. For precipitation hardening, the "curve" is the fundamental unit of data. The study by *Metals* (MDPI) on EN AW-6056 provides a high-fidelity kinetic dataset.[6]

4.1 Material and Experimental Parameters

- **Alloy:** EN AW-6056 (Al-Mg-Si-Cu).
- **Composition:** $\approx$ 0.94% Mg, 0.92% Si, plus Cu additions.

- **Experimental Grid:**
  - **Temperature (T):** 170°C, 180°C, 190°C.
  - **Time (t):** 1, 2, 3, 4, 6, 8, 12, 15, 18, 21, 24 hours.

4.2 Reconstructed Data: The Aging Matrix

The following table synthesizes the specific data points extracted from the research, mapping the aging conditions to mechanical outputs. This creates a "lookup table" for property prediction.

Table 1: Mechanical Property Evolution of AA6056 during Artificial Aging (Data derived from [6] and [6])

Aging Temp (°C)	Aging Time (h)	Condition State	Microhardness (HV0.5)	Yield Strength (MPa)	UTS (MPa)	Fracture Strain (%)
Baseline	0	T4 (Solution Treated)	83.2 ± 1.2	114.6 ± 8.9	242.6 ± 3.6	30.7
170	12	**Peak Hardness**	**148.9**	379.0 ± 15	428.9 ± 10	21.2
180	12	Over-Aged Transition	< 148	360.7 ± 13	406.1 ± 10	**25.2**
190	4	**Peak Strength**	~145	**400.5 ± 7**	**451.4 ± 4**	18.5

4.3 Second-Order Insights: The Hardness-Yield Strength Disconnect

A critical insight emerges from this dataset: **Maximum Hardness does not coincide with Maximum Yield Strength.**

- **Observation:** The sample aged at 170°C for 12 hours achieved the highest *Hardness* (148.9 HV), but the sample aged at 190°C for 4 hours achieved the highest *Yield Strength* (400.5 MPa).
- **Metallurgical Reasoning:**

- ○ **Hardness** (Indentation) is highly sensitive to the *number density* of precipitates. Lower temperatures (170°C) favor a higher nucleation density of very fine precipitates, resisting the local plastic deformation of the indenter.
- ○ **Yield Strength** (Macroscopic tension) is governed by the critical resolved shear stress needed to move dislocations through the entire grain. The higher temperature (190°C) likely promotes the formation of specific Cu-containing phases (L-phase or Q'-precursors) which act as more effective obstacles to dislocation glide on a global scale, or the slightly larger precipitates offer a different obstacle strength (Orowan vs. Shearing transition).

- **Implication for Modeling:** Models trained solely on hardness data (which is cheaper to generate) may underestimate the yield strength potential of alloys aged at slightly higher temperatures. Future datasets must distinguish between these two properties.

---

5. Kinetic Dataset Case Study: AA6014 (Pre-Aging & Bake Hardening)

For the automotive industry, the interaction between storage time (Natural Aging) and final curing (Paint Bake) is the single most important processing variable. The "Data in Brief" dataset by Yang et al. provides the gold standard for this specific domain.[11]

5.1 The Problem: Natural Aging (NA) Stability

Automotive sheets are solutionized by the supplier, shipped to the manufacturer (during which they naturally age), stamped, and then bake-hardened. The variability in "shipping/storage time" introduces noise into the final properties.

5.2 Dataset Structure

This dataset maps a multi-stage heat treatment process:

1. **Pre-Aging (PA):** T_{PA} (80–160°C) for t_{PA} (minutes to days).
2. **Natural Secondary Aging (NSA):** Storage at 20°C for time t_{NSA}.
3. **Paint Bake (PB):** Simulation of curing (180°C for 30 min).

Target Variable: Brinell Hardness (HBW).

5.3 Key Data Trends and Recommendations

The dataset reveals how to "immunize" the alloy against natural aging.

- **Unstabilized (No PA):** Hardness increases during NSA (cluster formation) but the final

PB response is poor. The clusters formed during NSA dissolve during the PB cycle rather than growing into β''.

- **Stabilized (With PA):**
  - **High PA Temp (160°C):** Promotes high PB response but lower NSA stability.
  - **Low PA Temp (80°C):** Promotes high NSA stability (hardness doesn't change during storage) but lower final PB response.
- **Optimal Window:** The data suggests an intermediate PA treatment (e.g., 100°C–120°C) balances these competing factors.

Mathematical Modeling Integration: The dataset explicitly tests kinetic models like the **Avrami (JMAK)** equation and the **Starink-Zahra** model.[11]

$$\alpha(t) = 1 - \exp(-(kt)^n)$$

Where α is the transformed fraction (related to hardness). The dataset provides fitted values for k (rate constant) and n (Avrami exponent), allowing researchers to simulate the hardening curves numerically.

---

6. Computational Materials Science: DFT and ML Generated Data

Beyond experimental measurements, the prompt asks for "dataset platforms" including computational sources. Two primary categories exist here: Ab Initio data and Machine Learning predictions.

6.1 Density Functional Theory (DFT) Datasets

Snippet [13] details a comprehensive DFT study of the Al-Mg-Si system.

- **Methodology:** Modeled 9 different supercells with varying Al, Mg, and Si atomic ratios using the Generalized Gradient Approximation (GGA).
- **Outputs:** Elastic constants (C_{11}, C_{12}, C_{44}), Bulk Modulus (B), Shear Modulus (G), Young's Modulus (E), and Poisson's Ratio (ν).
- **Shear Yield Strength Proxy:** The study calculates the **Shear Modulus (G)** to be **34.4 GPa** for the optimum Si/Mg ratio (4.5).
  - **Connection:** Since $\tau_{theoretical} \approx G/30$ to $G/10$, this fundamental parameter

allows for the estimation of theoretical shear strength limits.

  - **Pugh's Ratio:** The dataset includes Pugh's ratio (B/G), a metric for ductility. A ratio > 1.75 typically indicates ductility. The optimized Al-Mg-Si model showed a Pugh's ratio of **5.42**, confirming the high formability of these alloys.[13]

6.2 Machine Learning for Fatigue and Strength

The GitHub repository Fatigue-life-prediction-of-aluminum-alloy [14] and related papers [16] utilize datasets to predict complex failures.

- **Inputs:** Stress level, R-ratio, Composition, Yield Strength, UTS.
- **Algorithm:** XGBoost and Random Forest.
- **Data Integration:** These models often use static properties (YS, UTS) as *features* to predict *fatigue life* (N_f). The accuracy of these models relies heavily on the quality of the static property data discussed in Sections 3 and 4.
- **Key Finding:** Machine learning models have demonstrated the ability to predict the **0.2% Proof Stress** of Non-Isothermal Aging (NIA) schedules with high accuracy, identifying schedules that outperform standard isothermal aging.[18]

---

7. Synthesis: Data Availability and Gaps

7.1 Addressing the "Shear Yield Strength" Requirement

The user explicitly requested data on **Shear Yield Strength**. This is the most elusive property in standard datasets.

- **Findings:** Direct experimental datasets for Shear Yield Strength (τ_{yield}) in Al-Mg-Si are rare in open-access repositories. Standard tensile tests (ASTM E8) do not measure it.
- **Proxy Methods:**
  1. **Von Mises Calculation:** Most engineering datasets assume $\tau_{yield} = \frac{\sigma_{yield}}{\sqrt{3}} \approx 0.577\sigma_{yield}$. Given the extensive YS data in the "1154 Alloys" and "AA6056" datasets, this calculated value is the standard substitute.
  2. **DFT Derived:** As noted in Section 6.1, the calculated Shear Modulus provides a theoretical baseline.
  3. **Torsion Testing:** Specialized papers (not standard datasets) would be required for experimental values.

7.2 The 2xxx Series Comparison (High-Temperature Data)

While the focus is 6xxx (Al-Mg-Si), the creep dataset for **EN AW-2618A** (Al-Cu-Mg-Fe-Ni) [19] serves as a critical benchmark.

- **Relevance:** It provides a template for how high-temperature data should be structured: **Time vs Strain** at constant **Temperature/Stress**.
- **Gap:** A similar open-access creep dataset for Al-Mg-Si (e.g., AA6061 at 200°C) is currently missing from the major repositories (Mendeley, Kaggle). This represents a specific opportunity for future experimental work.

7.3 Data Formats and Interoperability

- **Kaggle/GitHub:** Data is often "flat" (CSVs with no metadata). These are useful for quick model testing but dangerous for rigorous metallurgy due to missing processing history.[21]
- **Data Journals (Data in Brief, Mendeley Data):** These provide the necessary metadata (solutionizing times, quench media) that determine the validity of the data. For professional application, these peer-reviewed datasets are the only reliable source.

---

8. Conclusion and Recommendations

The landscape of precipitation hardening data for the Al-Mg-Si ternary system has matured from isolated handbook values to structured, high-dimensional datasets capable of supporting advanced Machine Learning models.

Primary Data Sources for the Researcher:

1. **For Global Alloy Discovery:** The **1154 Alloys Dataset** (Mendeley Data, DOI: 10.17632/tvtg7gs59p.1) offers the widest breadth of compositional and processing descriptors.
2. **For Process Optimization (Kinetic Modeling):** The **AA6014 Pre-Aging Dataset** (Data in Brief, DOI: 10.1016/j.dib.2019.104494) and the **AA6056 Aging Matrix** (MDPI Metals) provide the high-resolution time-temperature series necessary to calibrate kinetic models.
3. **For Fundamental Physics:** DFT datasets (MDPI) provide the elastic constants and theoretical moduli that underpin the macroscopic mechanical behavior.

Key Metallurgical Takeaways:

- **Path Dependence:** Mechanical properties are not state functions of composition alone. The thermal path (Natural Aging $\rightarrow$ Pre-Aging $\rightarrow$ Artificial Aging) determines whether the alloy achieves high strength (via β'') or suffers from cluster-induced stunting.
- **Property Decoupling:** Hardness is an efficient proxy for screening but fails to capture the full yield strength potential, especially in Cu-bearing 6xxx alloys where complex phases (Q', L) evolve.
- **Shear Strength Estimation:** In the absence of direct shear datasets, the synthesis of Yield Strength data with Von Mises criteria or DFT-calculated Shear Modulus remains the

standard engineering approach.

For professional peers in alloy design, the integration of these disparate data sources—using the "1154 Alloys" dataset for coarse screening and the kinetic datasets for fine-tuning—offers a robust methodology for accelerating the development of next-generation Al-Mg-Si alloys.

---

9. References and Data Repository Access

Note: Citations are integrated into the analysis. Key identifiers for data access are summarized below.

- [22]
  Mendeley Data: "Aluminium Alloy Dataset for unsupervised learning", DOI: 10.17632/tvtg7gs59p.1.
- [12]
  Data in Brief: "Hardness data related to pre-ageing...", DOI: 10.1016/j.dib.2019.104494.
- [6]
  Metals (MDPI): "Effect of Artificial Aging Conditions on... EN AW 6056", DOI: 10.3390/met16010088.
- [21]
  Kaggle: "Aluminum-Yield Strength-Mg-Si-temp" (Amir Alizadeh).
- [19]
  Data in Brief: "Creep dataset for aluminum alloy EN AW-2618A...", DOI: 10.1016/j.dib.2025.112054.
- [13]
  Modeling: "Ab initio studies... of Al-Mg-Si alloys", DOI: 10.3390/constrmater2030015.

Works cited

1. Full article: Enhancing strength of Al-Mg-Si alloys through direct β′ phase precipitation induced by high-temperature cyclic deformation - Taylor & Francis, accessed on January 22, 2026, https://www.tandfonline.com/doi/full/10.1080/21663831.2025.2566252
2. Effect of Natural Aging on Precipitation Strengthening Behaviors in Al-Mg-Si Alloy - MDPI, accessed on January 22, 2026, https://www.mdpi.com/2075-4701/12/3/470
3. The Influence of Heat Aging Treatments on the Cavitation Erosion Behavior of a Type 6082 Aluminum Alloy - PMC - PubMed Central, accessed on January 22, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10488434/
4. Precipitation hardening in Al–Mg–Si alloys with and without excess Si - ResearchGate, accessed on January 22, 2026, https://www.researchgate.net/publication/257333069_Precipitation_hardening_in

_Al-Mg-Si_alloys_with_and_without_excess_Si
5. On the precipitation-hardening behavior of the Al-Mg-Si-Cu alloy AA6111 - ResearchGate, accessed on January 22, 2026, https://www.researchgate.net/publication/225622381_On_the_precipitation-hardening_behavior_of_the_Al-Mg-Si-Cu_alloy_AA6111
6. Effect of Artificial Aging Conditions on Mechanical Properties of EN ..., accessed on January 22, 2026, https://www.mdpi.com/2075-4701/16/1/88
7. Unsupervised machine learning discovers classes in aluminium alloys, accessed on January 22, 2026, https://royalsocietypublishing.org/rsos/article/10/2/220360/92096/Unsupervised-machine-learning-discovers-classes-in
8. Unsupervised machine learning discovers classes in aluminium alloys - PMC - NIH, accessed on January 22, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC9890099/
9. Phases observed during the precipitation hardening of Al-Mg-Si alloys - ResearchGate, accessed on January 22, 2026, https://www.researchgate.net/figure/Phases-observed-during-the-precipitation-hardening-of-Al-Mg-Si-alloys_tbl1_222171886
10. Inverse Design of Aluminium Alloys Using Genetic Algorithm: A Class-Based Workflow, accessed on January 22, 2026, https://www.mdpi.com/2075-4701/14/2/239
11. (PDF) Hardness data Related to pre-ageing, natural secondary ageing, and paint bake hardening in al-Mg-Si alloys - ResearchGate, accessed on January 22, 2026, https://www.researchgate.net/publication/335846263_Hardness_data_Related_to_pre-ageing_natural_secondary_ageing_and_paint_bake_hardening_in_al-Mg-Si_alloys
12. Hardness data related to pre-ageing, natural secondary ageing, and paint bake hardening in Al-Mg-Si alloys - PubMed, accessed on January 22, 2026, https://pubmed.ncbi.nlm.nih.gov/31673573/
13. Mechanical Properties of Al–Mg–Si Alloys (6xxx Series): A DFT-Based Study - MDPI, accessed on January 22, 2026, https://www.mdpi.com/2674-063X/2/3/15
14. Fatigue-life-prediction-of-aluminum-alloy/README.md at main - GitHub, accessed on January 22, 2026, https://github.com/1098994933/Fatigue-life-prediction-of-aluminum-alloy/blob/main/README.md
15. 1098994933/Fatigue-life-prediction-of-aluminum-alloy - GitHub, accessed on January 22, 2026, https://github.com/1098994933/Fatigue-life-prediction-of-aluminum-alloy
16. Application of machine learning algorithms to predict mechanical properties of aluminum alloys, accessed on January 22, 2026, https://pen.ius.edu.ba/index.php/pen/article/download/1278/902/2844
17. Machine Learning-Assisted Process Optimization of Al-Mg-Si Alloys - ResearchGate, accessed on January 22, 2026, https://www.researchgate.net/publication/390483710_Machine_Learning-Assisted_Process_Optimization_of_Al-Mg-Si_Alloys

18. Artificial intelligence inspired design of non-isothermal aging for γ–γ′ two-phase, Ni–Al alloys - NIH, accessed on January 22, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10403502/
19. Creep dataset for aluminum alloy EN AW-2618A in the T61 and overaged condition 1,000 h/190°C - ResearchGate, accessed on January 22, 2026, https://www.researchgate.net/publication/395605709_Creep_dataset_for_aluminum_alloy_EN_AW-2618A_in_the_T61_and_overaged_condition_1000_h190C
20. Creep dataset for aluminum alloy EN AW-2618A in the T61 and overaged condition 1000 h/190 °C - Fraunhofer-Publica, accessed on January 22, 2026, https://publica.fraunhofer.de/bitstreams/9409ea38-5943-44fa-8592-814056e56d7c/download
21. Aluminum-Yield Strength-Mg-Si-temp - Kaggle, accessed on January 22, 2026, https://www.kaggle.com/datasets/amiralizadeh9720/aluminum-yield-strength-mg-si-temp
22. Aluminium Alloy Dataset for unsupervised learning - Mendeley Data, accessed on January 22, 2026, https://data.mendeley.com/datasets/tvtg7gs59p/1